

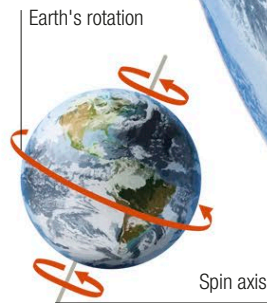
Earth

Our home is the largest of the inner planets, measuring 7,926 miles (12,756 km) across. Earth's surface has few visible impact craters—their remains have mostly been erased by the effects of air and water. The vast seas hide a unique feature: the ocean floor is being gradually replenished by fresh rock from long chains of underwater volcanoes. The sea floor moves very gradually like a giant conveyor belt, sinking back into the interior under the continents.



△ Water and life

Earth's distance from the Sun and the thickness of its atmosphere allow water to exist as a liquid, solid, and gas. The presence of liquid water is thought to have been essential for life to begin here. Why Earth has quite so much water is currently unknown.



△ Earth tilt and axial rotation

As Earth's spin axis is tilted by 23.5°, a part of it is always tilted toward the Sun. This gives rise to the seasons. During half of the year, the Northern Hemisphere receives most sunlight, and the Southern Hemisphere receives most during the other half.



Mars

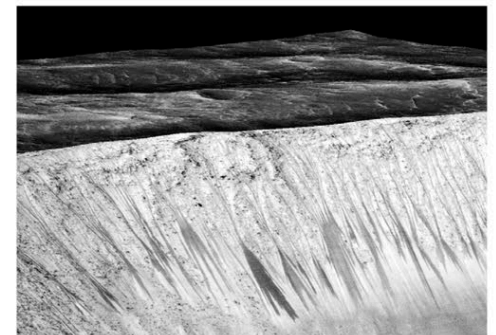
Mars has a surface almost equal in area to that of Earth's continents. The so-called Red Planet, at 4,220 miles- (6,792 km-) wide and most similar to Earth at its surface, is one of the few other places in our Solar System where life could have arisen, and may even exist today (see p.82). The planet's air is now too thin to support liquid water for long, even when the temperature creeps above 32°F (0°C). There is, however, plenty of evidence that the air was once thicker, and conditions for Martian life were much better billions of years ago.

Most the surface rocks on Mars have been oxidized, like the rust seen on some metals, giving them an orange colour.



△ Mars surface

Few worlds have terrains as diverse as Mars, from its heavily cratered ancient lands to smooth plains and plunging canyons. Its highest volcano, Olympus Mons, reaches 15 miles (25 km) above the planet's rocky surface, while the huge Valles Marineris canyon plunges 4 miles (7 km) below the surrounding plains.



△ Evidence of water

Mars displays features such as channels and canyons that clearly indicate that water flowed on its surface in the ancient past. There are also signs of brief water flows today, as seen on the slopes of this crater. Sunlight is thought to melt water ice buried under the surface, which flows before evaporating in the thin air.